least in Ruby), and they can respond to the messages they receive. But GenServers are not objects, and you should not treat them like objects.

It's important to consider that GenServers are a concurrency mechanism and objects are a programming development primitive. Their runtime performance characteristics are also completely different. You can create and destroy many more objects per second in Ruby than you could processes in Elixir—simply because Ruby objects are lighter weight and are designed to be created and destroyed rapidly.

Another aspect that GenServers handle that objects don't is time. Every program that we write runs over a period of time. Often, we want to do things based on a certain frequency. An object in Ruby lacks the ability to easily deal with time. We would need to build that ourselves and deal with all of the challenges that would be introduced. However, we can easily create a GenServer that changes over time by sending messages in an interval.

This time aspect lets you think of GenServers as a living thing. They can change over time (in the constraints that are coded for them). They emit messages to other GenServers in the system. They can be entirely contained or very sociable. They can exist for the duration of a web request, a WebSocket connection, or even exist as long as the system is up. Thinking of them in this way lets you build things that would be considered complex but are intuitive to develop and reason about.

Now that you know how to create a GenServer, let's step back and understand some risks that come with them.

What About Supervisors?



Supervisors are an important type of process in BEAM applications. A Supervisor keeps track of child processes and can strategically restart them if one dies. You'll see a type of Supervisor later in this book when we discuss the Application module.

This book doesn't cover Supervisors because you don't need to know much about them this early in your Elixir journey. As you learn Elixir and get more comfortable, you'll likely end up using them. When that happens, read the Supervisor documentation.[5]

5. https://hexdocs.pm/elixir/1.12/Supervisor.html